

Revision Diff Report: v1 → v2

Volatility Absorption: The Diminishing Marginal Impact of Market Fear

Date: 2026-03-30

Issues Addressed: L1 (HIGH), S1 (HIGH), R1 (HIGH)

Summary of Changes

Metric	v1	v2
Total pages	33	38
Bibliography entries	21	37
New references added	—	16
Sections restructured	—	2 (Methodology, Results)
New paragraphs added	—	4
Net new text	—	~1,400 words

Issue L1 (HIGH): NSI Mechanical Correlation Undermines Identification

Reviewer concern: The primary measure $NSI = |r_t|/V_t$ mechanically declines with V_t because VIX is in the denominator AND defines the conditioning regimes.

Changes Made

- [Δ] Abstract: Reframed to present SAR as primary measure.
 - [–] ~~“Our central measure is the normalized shock impact—the absolute return on a shock day divided by the prevailing VIX level.”~~
 - [+] “we measure absorption primarily through the *shock amplification ratio* (SAR)—the ratio of shock-day to normal-day absolute returns within each VIX regime—which avoids the mechanical denominator problem inherent in VIX-normalized measures.”
 - [+] Note about NFP limitation added to abstract.
- [Δ] Introduction §1: SAR promoted to primary, NSI demoted to supplementary.
 - [–] ~~“Our central measure is the normalized shock impact...”~~
 - [+] “Our primary measure is the *shock amplification ratio* (SAR)—the ratio of mean absolute returns on shock days to non-shock days *within* each VIX regime—which is free of the mechanical denominator problems...”
 - [+] “As a supplementary continuous-variable summary, we also compute the *normalized shock impact* ($NSI = |r_t|/V_t$)... We emphasize that this ‘paralysis’ coefficient is consistent with—but does not substitute for—the SAR evidence, because the NSI construction introduces a mechanical negative correlation through the VIX denominator.”
- [Δ] Methodology §3: Major restructuring of subsections.

- ~~[-] Subsection “Normalized Shock Impact” was the first methodology subsection.~~
 - [+] New ordering: (1) VIX Regime Classification, (2) **Shock Amplification Ratio (Primary Measure)**, (3) Normalized Shock Impact (Supplementary Measure).
 - [+] SAR subsection explicitly states: “Crucially, the SAR compares shock-day to non-shock-day absolute returns *within* each VIX bin and does not divide by VIX. It is therefore free of the mechanical denominator problem.”
 - [+] New paragraph “**Mechanical correlation caveat**” added to the NSI subsection, explicitly explaining the mechanical correlation problem and three pieces of mitigating evidence.
4. [Δ] Results §5.1: SAR analysis explicitly labeled as primary identification evidence.
- [+] “Because the SAR does not divide by VIX, it avoids the mechanical denominator problem inherent in VIX-normalized measures and constitutes our primary identification evidence.”
 - [Δ] NSI regression relabeled as “supplementary” with caveat: “We reiterate, however, that the NSI regression carries a mechanical component from the VIX denominator (see Section 3), and the SAR analysis above provides the cleaner identification.”
5. [Δ] Limitations §8.1: Strengthened acknowledgment of mechanical correlation.
- ~~[-] “our central measure—the normalized shock impact...”~~
 - [+] “the supplementary normalized shock impact measure ($NSI_t = |r_t|/V_t$) uses VIX in the denominator...”
 - [+] “For this reason, we designate the SAR—which does *not* divide by VIX—as our primary identification measure.”
 - [+] Reference to Patton (2011) as alternative methodology for future work.
6. [Δ] Future Research §8.2: Added simulation exercise suggestion.
- [+] “simulation exercises quantifying the mechanical component of the NSI regression—for example, generating returns under a null of no absorption and measuring the spurious slope attributable to the VIX denominator—would further sharpen identification; the forecast comparison methodology of Patton (2011) offers an alternative approach.”

Issue S1 (HIGH): NFP Event Study Lacks Surprise Magnitude Control

Reviewer concern: The paper uses NFP day dummy without controlling for surprise component (actual minus consensus). The declining NFP effect could reflect smaller surprises during high-VIX periods.

Changes Made

1. [Δ] Abstract: Added limitation note.
 - ~~[-] Specific NFP numbers in abstract ($1.24\times$, $1.30\times$, $0.95\times$).~~
 - [+] “Event-level evidence from nonfarm payroll releases is directionally consistent, though we note the limitation of using a binary NFP dummy rather than surprise-magnitude controls.”
2. [Δ] Methodology §3.5: Pre-emptive caveat added.

- [+] “We note that the standard approach in this literature uses the *surprise component* (actual minus consensus forecast) as the key regressor; our analysis uses a binary NFP dummy due to data constraints, and we discuss the implications of this limitation in Section 5.”
- [+] Citations: Andersen et al. (2003), Balduzzi et al. (2001).

3. [+] Results §5.3: New paragraph “**Limitation: absence of surprise control**” (~200 words).

- Explains the alternative interpretation: declining NFP effect could reflect smaller surprises during high-VIX periods.
- References Andersen et al. (2003) and Balduzzi et al. (2001) methodology.
- Notes that NFP evidence is “supplementary to the SAR-based primary analysis.”
- Provides the correct specification for future work: $|r_t^{SPY}| = \alpha + \beta_1|\text{surprise}_t| + \beta_2V_t + \beta_3(|\text{surprise}_t| \times V_t) + \varepsilon_t$.
- Identifies data sources: Bloomberg, Federal Reserve Bank of Philadelphia real-time dataset.

4. [Δ] Future Research §8.2: Added NFP surprise extension.

- [+] “the NFP event study could be strengthened substantially by incorporating surprise-magnitude controls following Andersen et al. (2003), using consensus forecast data from Bloomberg or the Federal Reserve Bank of Philadelphia’s real-time dataset.”

Issue R1 (HIGH): Missing Critical References

Reviewer concern: Only 20 references; JBF norm is 40–60. Missing key citations on regime-dependent volatility, news impact curves, attention, and VRP.

New References Added (15)

#	Reference	Journal	Where Cited in v2
1	Andersen, Bollerslev, Diebold & Vega (2003)	AER	§3.5 (NFP methodology), §5.3 (limitation caveat), §8.2 (future work)
2	Balduzzi, Elton & Green (2001)	JFQA	§3.5 (NFP methodology), §5.3 (limitation caveat)
3	Baur & Lucey (2010)	Financial Review	§1 (GLD as safe haven)
4	Bekaert, Engstrom & Xu (2022)	Management Science	§2.2 (VRP time variation), §5.5 (VRP results)
5	Chernov, Lochstoer & Lundebry (2022)	RFS	§2.2 (risk-return trade-off)
6	Da, Engelberg & Gao (2015)	RFS	§1 (intro lit connections), §2.4 (FEARS index)
7	Danielsson, Valenzuela & Zer (2018)	RFS	§1 (endo/exo risk), §5.4 (endo/exo results)
8	Drechsler & Yaron (2011)	RFS	§2.2 (uncertainty premia), §5.5 (VRP results)
9	Engle & Ng (1993)	JoF	§1 (intro lit), §2.1 (news impact curves)

#	Reference	Journal	Where Cited in v2
10	Martin (2017)	QJE	§2.2 (expected returns)
11	Muler & Yohai (2008)	JSPI	§2.1 (robust GARCH)
12	Patton (2011)	J. Econometrics	§8.1 (limitations), §8.2 (future work)
13	Todorov (2010)	RFS	§2.2 (VRP and jumps), §5.5 (VRP results)
14	Vlastakis & Markellos (2012)	JBF	§2.4 (information demand), §5.4 (endo/exo)
15	Whaley (2000)	JPM	§1 (intro lit), §5.4 (fear gauge)
	Zakoian (1994)	JTSA	§1 (intro lit), §2.1 (TGARCH)

Note: 16 new entries total. Zakoian (1994) was listed separately in the review but is included in the additions above.

Text Integration Points

1. [Δ] Literature Review §2.1 (GARCH): Substantially expanded.

- [+] Zakoian (1994) TGARCH as closest parametric analogue to absorption.
- [+] Engle & Ng (1993) news impact curve framework as natural comparison.
- [+] Muler & Yohai (2008) for robust estimation under heavy-tailed outliers.
- [+] Explicit differentiation: our contribution documents declining marginal impact *within* high-volatility regime, which standard regime-switching models do not capture.

2. [Δ] Literature Review §2.2 (VRP): Substantially expanded.

- [+] Todorov (2010) on jump risk premium.
- [+] Drechsler & Yaron (2011) on macroeconomic uncertainty premia.
- [+] Bekaert, Engstrom & Xu (2022) on time-varying risk appetite.
- [+] Martin (2017) on alternative expected return framework.

3. [Δ] Literature Review §2.4 (Behavioral): Substantially rewritten.

- ~~[-] Misattribution of “panic fatigue” / “crisis habituation” to Kahneman & Tversky (1979) and Barberis et al.~~
- [+] Da et al. (2015) FEARS index as direct fear measure.
- [+] Vlastakis & Markellos (2012) information demand framework.
- [+] Careful separation: prospect theory provides general foundation for diminishing sensitivity, but absorption is more directly linked to attention-saturation models.
- [+] Explicit statement: “we avoid attributing it to prospect theory directly.”

4. [Δ] VRP Results §5.5: Added theoretical context.

- [+] “consistent with the theoretical frameworks of Todorov (2010) and Drechsler and Yaron (2011)”
- [+] Bekaert et al. (2022) citation for time-varying risk appetite.

5. [Δ] Introduction §1: Added 5 new citations throughout.

Reference Count

	Count
v1 bibliography entries	21
New entries added in v2	16
v2 total bibliography entries	37
JBF typical range	40–60

The reference list has grown from 21 to 37 entries. While still below the JBF median (~ 50), the additions address all specific gaps flagged in the review (regime-dependent volatility, news impact curves, investor attention/fear, VRP dynamics, macro event studies). Further references can be added during subsequent revision rounds as additional robustness analyses are completed.

Additional Minor Fixes (Bonus)

1. ~~[-] \bibliographystyle{apalike} — removed because the bibliography is manual (the bibliography environment), making the style declaration a no-op.~~ (Review FMT2)
2. [Δ] Behavioral citations in §2.4: Fixed misattribution of “panic fatigue” to prospect theory. (Review R2, citation_check P5)
3. [Δ] VRP discussion: Now cites specific theoretical frameworks rather than vague “some prior claims.” (Review R3, citation_check P4)

What Was NOT Changed (Preserved for Future Rounds)

The following MEDIUM/LOW issues from review_v1 were **not** addressed in this revision and remain open for future rounds:

- L2: SAR non-monotonicity in crisis regime (needs structural break test)
- M1 (Methodology): Shock-type GLD threshold sensitivity analysis
- C1: Sharper differentiation from regime-switching literature
- T1: Adding figures (bar charts, scatter plots)
- S2: Multiple comparisons correction
- L4: Proposition relabeled to Remark
- Additional 5–15 references to reach JBF median of ~ 50